



LEE PIN LOON

Robotics Software Engineer

CONTACT DETAILS

✉ pinloon_0428@hotmail.com

SKILLS

- C++ 11/14/17/20
- Embedded C programming
- Robotics (mapping & planning)
- Python
- ROS1/ROS2

PROFILE LINKS

- 🐙 github.com/pllee4
- 📖 pinloon.medium.com
- 🌐 linkedin.com/in/pinloonlee
- 📖 pinloon.gitbook.io
- 📺 youtube.com/@pllee4

ABOUT ME

- Meticulous in code review
- Code hobby projects
- Play badminton

LANGUAGE

English, Mandarin, Hokkien, Malay

WORKING EXPERIENCE

Dyson Operations Pte Ltd

June 2022 – Present

- ◇ Domain owner for robot mapping and global path planning
- ◇ Develop in-house open-area and edge-cleaning pattern for D-shaped vacuum cleaning robot
- ◇ Implement multi-attempt replanning algorithm for reliability
- ◇ Develop autozoning algorithm for room segmentation
- ◇ Develop multi-map handling and switching for different scenarios
- ◇ Implement user map editing and restriction zones
- ◇ Modularize planning algorithm to support different strategies
- ◇ Optimize code to improve speed and memory usage

Weston Robot Pte Ltd

Oct 2020 – May 2022

- ◇ Robotics engineer specialized in firmware (RTOS) and algorithm
- ◇ Supported development of SDK in C++ for mobile robots
- ◇ Developed in-house obstacle avoidance based on ultrasonic sensors
- ◇ Developed safety control logic on firmware for mobile robots
- ◇ Supported firmware with POSIX on Linux for fast testing
- ◇ Developed in-house communication protocol for data exchange
- ◇ Designed handshaking algorithm between core controller and user
- ◇ Developed drivers for data acquisition of sensors and actuators
- ◇ Improved performance and reliability of mask detection robot by having self-recovery, smart alert notification and data logging

ACHIEVEMENTS

- ◇ 1st prize in Hack for Good 2.0 Hackathon 2019
- ◇ 1st runner up and Crowd Favourite in The Great Lab (TGL) Grand Design Challenge 2019
- ◇ Best Idea Award in ABU ROBOCON Vietnam 2018
- ◇ Champion and 1st Runner-Up with Best Engineering Award in ROBOCON MALAYSIA 2018
- ◇ 1st Runner-Up in ABU ROBOCON Tokyo 2017
- ◇ Referee in International Youth Robot Competition (IYRC) 2017

ACTIVITIES

Traffic Management Project with CIQ complex

2020

- ◇ Traffic flow visualisation with LSTM prediction model

Water Quality IoT System

2019

- ◇ Handled embedded programming to have real time water quality monitoring system of river

Technical Project with Data Engineer

2018

- ◇ Database creation and visualisation for agricultural information using Python and SQLite

UTM ROBOCON Team

2016 – 2018

- ◇ Robot programmer with 2 years of experiences, handling robot program for 2 robots in ROBOCON Malaysia
- ◇ Operator and robotic programmer of Malaysia Team in ABU ROBOCON Vietnam 2018
- ◇ Developed implementation of Fuzzy PID controller on robots

EDUCATION

National University of Singapore

Oct 2023 – June 2025

- ◇ Master of Science in Robotics

GPA: 4.53 / 5.0

University Technology Malaysia

Sep 2016 – June 2020

- ◇ Bachelor of Engineering (Electrical - Mechatronics)

CGPA: 3.94 / 4.0